



CHAPTER

Maintenance and troubleshooting overview

General schedule and fault-finding approach — for a specific compartment’s checks, specifications and fault table, see its own chapter in Part II.

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7.1 Safety during maintenance

Before performing any wiring or sensor checks, isolate and verify the control circuit: control voltage is referenced to ground at 24 V DC — confirm 24 V DC is present between the terminal and PE before assuming a wiring fault, and never connect a sensor to the PLC before bench-testing it (chapter 13). TODO: no dedicated safety document (lockout/tagout, PPE, chemical/electrode handling) specific to SalinBloc™ maintenance was found — see chapter 2.

7.2 Maintenance schedule

No complete maintenance interval table (daily/weekly/monthly/annual) was found for the SalinBloc™ system as a whole. The one concrete, recurring interval documented anywhere in HydroVolta's engineering library is the CIP (clean-in-place) cycle described in chapter 12, which ran approximately **weekly** in the documented pilot; the pretreatment UF stage is otherwise cleaned on a TMP-triggered basis rather than a fixed interval (chapter 8).



NOTICE

The table below is a **recommended starting schedule**, not a HydroVolta-issued one. The two rows marked "Confirmed" come directly from source documents cited elsewhere in this manual; every other row is this manual's synthesis of general good practice for equipment of this type (visual/level checks, periodic sensor verification, periodic actuator exercise, annual inspection), pending review and correction by HydroVolta engineering once real field data exists. Do not treat the unconfirmed rows as validated intervals.

Table 7.1. Recommended starting maintenance schedule

Task	Interval	Source / basis
Visual walk-round: check for leaks, unusual noise/vibration, tank levels, HMI alarm state	Daily	Recommended practice (unconfirmed)
Record process values in maintenance log (appendix A ranges)	Daily	Recommended practice (unconfirmed)
CIP cycle (acid + hypochlorite steps, chapter 12)	Weekly	Confirmed — documented pilot interval
Check dosing chemical tank levels and top up	Weekly	Recommended practice (unconfirmed)
UF chemically enhanced backwash (CEB)	When TMP rises ≥ 0.9 bar above initial	Confirmed — UF manufacturer spec (chapter 8)
Bench-verify each 4–20 mA sensor loop (section 13.4)	Quarterly, or on any suspect reading	Recommended practice (unconfirmed)
Exercise/inspect valve actuators (manual override, seat wear)	Quarterly	Recommended practice (unconfirmed)
Inspect control cabinet (dust ingress on IP55 enclosure, terminal torque)	Semi-annually	Recommended practice (unconfirmed)
Full system inspection and log review	Annually	Recommended practice (unconfirmed)

TODO: every compartment's *Maintenance* section in Part II should be reviewed by HydroVolta engineering and this table corrected/expanded with real intervals as field data accumulates.

7.3 Maintenance log

A maintenance log should record, per event: date, technician, equipment/asset ID (e.g. stack, pump, sensor tag), the check or service performed, readings taken (pressure, flow, pH, conductivity, etc.) versus expected range, parts replaced (with part number), and any follow-up action required. Its purpose is a traceable per-unit service history so recurring faults, consumable replacement intervals and warranty-relevant events can be reviewed over the system’s lifetime.

Table 7.3. Maintenance log template

Date	Technician	Equipment / tag	Check / service performed	Result

7.4 Alarm handling

Per HydroVolta's HMI design specification (chapter 6), alarms follow the ISA-18.2 alarm-management convention and always resolve the same way, regardless of which alarm fired:

1. The PLC raises the alarm; it appears on the **Alarm Summary** screen with its code, severity, cause, effect, timestamp and a recommended action.
2. The operator investigates using that information and this manual's Troubleshooting section for the relevant compartment (below).
3. Once the operator has acknowledged they are aware of it, press **ACK Alarm** (Mode Control screen) — the alarm moves to the acknowledged list.
4. Physically resolve the underlying condition.
5. Press **Reset Trip** — only enabled once the condition has actually cleared, so a reset that won't take is itself informative: it means the underlying condition has not actually gone away yet.
6. Resolved alarms remain in the alarm *history* with their full timestamp trail, for the maintenance log (chapter 7).

A fault severe enough to trip **Safe Stop** needs the fuller recovery sequence in section 6.9, not just ACK + Reset.

7.5 How to use the troubleshooting sections

Each Part II chapter ends with a *Troubleshooting* section scoped to that compartment: symptom, likely cause, and remedy. Start with the compartment most closely tied to the symptom (e.g. a conductivity alarm → chapter 13 or the relevant tank chapter (chapters 15 to 18); a communication timeout → chapter 14 or chapter 19). If the fault spans several compartments or persists after following the relevant section, escalate per section 7.6.

7.6 When to call service

Escalate to HydroVolta service when: a sensor fails its bench test (chapter 13) and no spare is available on site; a remote-access device (e.g. the ADAM-3600 RTU) remains unreachable after the documented recovery steps (chapter 14) and is not even pingable; or process readings remain outside the documented operating envelope after checking feedwater quality, pump duty point and pipe sizing. TODO: no source document defines formal escalation criteria, response-time commitments, or service contact information — see also chapter 1 (Contact and support).

